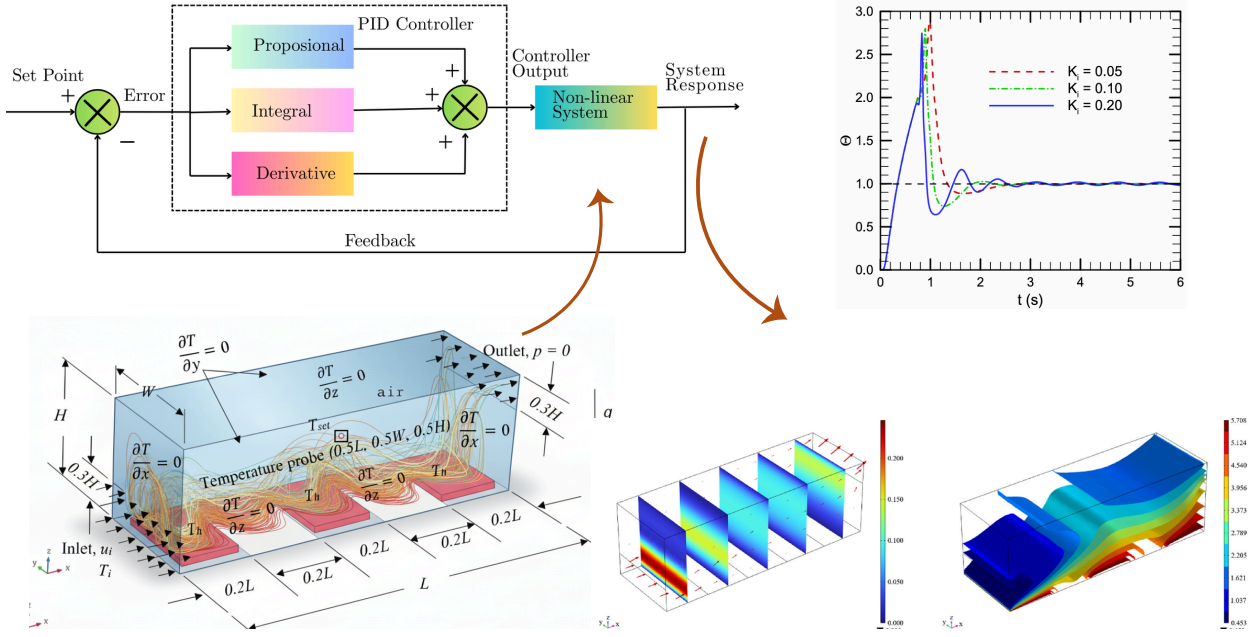


Thermal Management and Robustness of Linear Controllers (P, PI, and PID) in Mixed Convection - A CFD Approach

Employing the *control law* for regulating *convection* can provide an optimal system for many sophisticated applications, such as MEMS fabrication, electronic component cooling, and drug manufacturing facilities. A *computational approach* is always advantageous in this regard for affordable trial and error. A more advanced version of this can be established by creating a *digital twin* for the application. Other improvements include the use of *nonlinear controllers* for the system, inserting multiple sensors in the domain, and acquiring precise measurements. Another aspect involves *quantifying the uncertainty* and using it during feedback. This would require *advanced data analysis* in real-time and *statistical methods* to capture the system's dynamics and enable precise control.

A related published work (Deb et al., 2023[†]) only touches upon the basic aspects. This can be leveraged for many advanced applications with the aforementioned improvements.

[†]Deb, N., Fardin, S., Fardin, M. M., Nawal, N., Nizam, M. R., & Saha, S. (2023). Thermal management inside a discretely heated rectangular cuboid using P, PI and PID controllers. Case Studies in Thermal Engineering , 51, 103601. <https://doi.org/10.1016/j.csite.2023.103601>



Continuity:	$\nabla \cdot \mathbf{u} = 0$
Momentum:	$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} \right) = -\nabla p + \mu \nabla^2 \mathbf{u} + \rho \mathbf{g} \beta (T - T_0)$
Energy:	$\rho c_p \left(\frac{\partial T}{\partial t} + \mathbf{u} \cdot \nabla T \right) = k \nabla^2 T$
Control law (PID):	$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt} \quad e(t) = T_p - T_{set}$